

On Refining Polynomial Phase Signal Parameter Estimates

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Newton algorithms are commonly used in the final “refinement” stage of parameter estimation for sinusoids and higher order phase polynomial signals. Such parameter estimation scenarios arise commonly in radar applications where the radial target velocity (which is possibly time-varying) must be estimated. The author and coworkers have previously proposed an elegant and efficient alternative to a Newton gradient search type algorithm based on filtering and phase unwrapping. A statistical and computational analysis of this filtering/phase unwrapping method is presented here. The analysis shows the algorithm to be computationally efficient and much less sensitive to the accuracy of the initial guesses for the parameters than a Newton algorithm. A first-order statistical analysis of the filtering/phase unwrapping algorithm is performed, and guidelines are derived for the required accuracy of the initial estimates. Simulations are presented to confirm the analysis.

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I. INTRODUCTION

Estimation of frequency from discrete-time observations is an important practical problem. In radar, for example, Doppler frequency estimation enables estimates of radial target velocity to be obtained. The optimal maximum-likelihood (ML) procedure for estimating frequency from sinusoids involves maximizing a discrete-time Fourier transform of the observation. In practice, the estimation is typically done in two stages. In the first stage, a fast Fourier transform (FFT) is applied to the signal, and a “coarse” search is performed over the “bins” of the FFT [1]. This first stage yields an initial estimate for the frequency. In the second stage, a Newton algorithm is typically used in conjunction with a simple interpolation formula to yield an optimal final estimate. In many radar scenarios, the target's radial velocity is not constant, and it is necessary to account for this to obtain accurate velocity estimates. In [2], for example, the velocity was modeled as a first-order polynomial in time. The estimation procedure in [2] used an initial coarse two-dimensional (2D) search, followed by a 2D gradient search. In subsequent work, higher order polynomial laws were assumed for the variation in the velocity, and some new suboptimal algorithms were proposed for obtaining the initial polynomial phase parameter estimates [3, 4]. Once initial estimates are obtained via the methods in [3, 4], one can use Newton algorithms for refining the suboptimal estimates to the point of asymptotic optimality.

Parameter estimation of polynomial phase signals is not only important in conventional radar applications, but in synthetic aperture radar (SAR) as well. In SAR imaging, the transmitted signal often has a time-varying phase shift arising from the relative motion between the transmitter and target. Polynomial phase parameter estimation can be used to deal with this complication, as discussed in [5].

The refinement algorithm considered in this paper can be divided into four steps: 1) de-chirping the observation to “baseband” (using coarse initial estimates for the polynomial phase parameters), 2) applying simple M -point moving average filtering followed by M -fold decimation, 3) polynomial phase estimation of the de-chirped, filtered, and decimated signal via phase unwrapping and least-squares estimation, and 4) appropriately combining the estimates in stage 3 with the initial coarse estimates to obtain the final estimates. This four stage process was briefly outlined in [12] and [11], but no statistical analysis was performed, and no guidelines were presented for the required accuracy of the “initial guesses.” A method for refinement of polynomial phase parameters based on least-squares phase unwrapping was also presented in [8], but it did not include the low-pass filtering/decimation steps.

Section II presents the refinement algorithm. Section III considers the computational and statistical performance of the algorithm, while Section IV discusses the application of the new algorithm and provides simulation examples. The conclusions are presented in Section V. The Appendix contains derivations of the statistical properties of the estimators and investigates how close the initial guesses for the parameter estimates need to be.

II. THE REFINEMENT ALGORITHM

A. Signal Model and Algorithm

The discrete-time polynomial phase signals considered in this paper can be represented as

$$z(n) = z_s(n) + z_w(n) = b_0 \exp \left\{ j \sum_{k=0}^P a_k n^k \right\} + z_w(n), \quad |n| \leq \frac{N-1}{2} \quad (1)$$

where b_0 is the signal amplitude, P is the polynomial phase order, the $\{a_k, k = 0, \dots, P\}$ are the phase parameters to be estimated, $z_s(n)$ is the signal component of the observation, $z_w(n)$ is additive complex white noise, and N is the record length (an odd integer). It is assumed that the noise has a power of σ^2 .

To reduce ambiguities that occur in phase unwrapping, it is assumed that the a_k are constrained such that the change in phase between any two consecutive samples is less than π in magnitude. A mathematical formula to express the aforementioned constraint on the change of phase between consecutive samples is

$$\left| \sum_{k=1}^P a_k (n^k - n^{k-1}) \right| < \pi, \quad |n| \leq \frac{N-1}{2}. \quad (2)$$

The filtering/phase unwrapping algorithm is specified with the 4-step procedure below.

1) Determine coarse initial estimates, $\{\hat{a}_{k_i}, k = 1, \dots, P\}$, for the polynomial phase parameters. Then use these initial parameter estimates to de-chirp the observed signal:

$$z_d(n) = z(n) \exp \left(-j \sum_{k=1}^P \hat{a}_{k_i} n^k \right). \quad (3)$$

At the end of this step (assuming the coarse estimation has been performed adequately), the signal content will be highly localized around the 0 Hz spectral region. The noise content, on the other hand, will be evenly dispersed across the entire spectral band.

2) Lowpass filter the de-chirped signal with a simple moving average filter, and decimate:

$$z_o(m) = \frac{1}{M} \sum_{n=mM-(M-1)/2}^{mM+(M-1)/2} z_d(n), \quad m = -Q, -Q+1, \dots, +Q \quad (4)$$

where $Q = \lfloor (\frac{1}{2}((N/M) - 1)) \rfloor$ and M is the number of samples in the moving average filter impulse response. M should be small compared with N , and it must be odd. The filtering in this step is used to increase the signal-to-noise ratio (SNR). Assuming adequate coarse initial estimates have been obtained, this occurs because most of the noise content falls outside the passband of the filter, whereas the signal content falls substantially within the passband of the filter. As explained in the Appendix, the filtering tends to cause the SNR to increase by a factor of approximately M . Decimation (by a factor of M) is possible because the filtering reduces the effective bandwidth of the signal. The increased SNR afforded by the filtering enables phase unwrapping and linear least-squares estimation techniques to be used. These are the basis of the next two steps.

3) Create a vector, $\mathbf{V}$, of unwrapped phase estimates: $\mathbf{V} = \{V(m), m = -Q, \dots, Q\} = \text{unwrap}(z_o(m))$, where $\text{unwrap}(\cdot)$ indicates that the angle of $(\cdot)$ is unwrapped. $\mathbf{V}$ is then modeled as a new polynomial phase signal in noise with unknown phase parameters, $\mathbf{a}_d = [a_0 \ \delta a_1 \ \dots \ \delta a_P]^T$, where $\delta a_k = a_k - \hat{a}_{k_i}$, $k = 1, \dots, P$. These new phase parameters are estimated according to

$$\hat{\mathbf{a}}_d = [\hat{a}_0 \ \hat{\delta a}_1 \ \dots \ \hat{\delta a}_P]^T = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \mathbf{V} \quad (5)$$

where

$$\mathbf{G} = \begin{bmatrix} 1 & -Q & \dots & (-Q)^P \\ 1 & -Q+1 & \dots & (-Q+1)^P \\ \dots & \dots & \dots & \dots \\ 1 & Q & \dots & Q^P \end{bmatrix}. \quad (6)$$

4) Obtain final estimates for the phase parameters using the relations:

$$\hat{a}_{k_f} = \hat{a}_{k_i} + \hat{\delta a}_k / M^k, \quad k = 1, \dots, P \quad (7)$$

$$\hat{a}_{0_f} = \hat{a}_{0_i}. \quad (8)$$

Note that in some applications, the polynomial phase order will not be known a priori. In such cases, one can start by analyzing the observation based on a comparatively low order and then repeat the analysis with successively higher model orders. The power of the residual (i.e., the power of the error between the estimated and observed signals) provides some indication of when the appropriate model order has been reached—the residual power will stabilize. In choosing the “best” order, one has to deal with the tradeoff between quality of fit and the

complexity of the model. That is, one has to choose a “parsimonious” model. The issue of model order selection for polynomial phase signals is addressed in detail in [13].

In general, polynomial curve fitting is ill-conditioned if the number of samples in the observation or the polynomial order are very large, so the process is vulnerable to round-off errors. Fortunately, for the polynomial curve fitting in (5) and (6), the number of effective samples has been reduced by the preceding decimation process. If the number of samples does become large (even after decimation), or the polynomial order is large, the polynomial curve fitting procedure defined in (5) and (6) could be replaced with a more numerically robust orthogonal polynomial curve-fitting technique [15].

III. STATISTICAL AND COMPUTATIONAL PERFORMANCE OF THE ALGORITHM

Step 3 of the filtering/phase unwrapping algorithm uses a phase unwrapping/linear regression technique. There are a number of alternative ways of implementing the phase unwrapping [6, 7]. Regardless of the method, the phase unwrapping is only effective above a certain SNR = S_t , where $\text{SNR} = b_0^2/\sigma^2$. For all examples in this paper, the phase unwrapping approach in [6] is used and $S_t \approx 9$ dB. Fortunately, the step in the algorithm prior to the phase unwrapping is a filtering operation that provides an M -fold enhancement in SNR. This enhancement occurs (as discussed in the Appendix) provided that the initial coarse estimates are sufficiently close, i.e.,

$$|\delta a_k| < \frac{\pi}{4kPM(N/2)^{k-1}}, \quad k = 1, \dots, P \quad (9)$$

where $\{\delta a_k, k = 1, 2, \dots, P\}$ is the error in the coarse initial estimate for a_k . Assuming that the M -fold SNR enhancement does occur, then the SNR threshold of the input (i.e., the observation) needs to be

$$\text{SNR}_{\text{input}} > \frac{S_t}{M}. \quad (10)$$

It is important to make some comments about the effect of the filtering operation. First, the filter is linear phase and hence introduces no phase distortion. Second, the de-chirped signal should be concentrated around the 0 Hz region, where the sinc-style transfer function is fairly flat. There should also, therefore, be limited amplitude distortion. The small amount of distortion arising from the nonflat passband can be limited by making the filter passband especially broad, or equivalently by making M especially small. M should therefore be made as small as possible without violating requirement (10), i.e., without violating the relationship

$$M > \frac{S_t}{\text{SNR}_{\text{input}}}. \quad (11)$$

The above equation can be used to determine the required value of M if the input SNR is known or

can be reliably estimated. An additional way to limit the problem of amplitude distortion is to perform a second iteration of the algorithm in which the estimates obtained after Step 4 of the algorithm are used as the new “coarse estimates,” and the algorithm is reapplied.

It is of interest to know how to choose M if the SNR of the input is not known. In this case, one can start by choosing a small value of M and then using the filtering/phase unwrapping method to estimate the signal parameters. There are a number of ways of gauging whether estimation with the current value of M has been successful. One can, for example, de-chirp the observation by all the estimated phase parameters and then take the discrete Fourier transform (DFT); if the peak of the DFT is close to dc (i.e., within a few multiples of the Cramer-Rao (CR) bound for the a_1 parameter), then the estimation can be deemed successful. An alternative approach, which is preferred because it is computationally less intensive, involves de-chirping the signal to baseband once the final parameter estimates are computed to obtain $z_d(n)$. If the estimation process has been successful, $z_d(m)$ will be approximately a dc signal imbedded in noise with an SNR greater than about $S_o = 9 - 10\log_{10} M$ dB, i.e., it will have an SNR that permits effective phase unwrapping. If the estimation has not been successful, then one can iteratively increase the value of M until the estimation is successful. At extremely low SNRs, it may be that the method will never be successful.

A first-order statistical analysis of the algorithm is performed in the Appendix, which shows that if $N \gg M$, the parameter estimate variances approach the CR bound as $N \rightarrow \infty$.

Now consider the computations required for the algorithm. For Step 1 of the algorithm in Section II, the de-chirping operation requires approximately N complex multiplications (with each complex multiplication comprising four real multiplications and two real additions), PN real multiplications, PN real additions, and PN “exponent” operations. Note that an exponent operation pertains to the raising of a number to some power, and can be done with about 15 real multiplications, a similar number of additions and 7 stored constants [14]. For Step 2, the filtering requires approximately N complex additions. For Step 3, the phase unwrapping procedure requires $\approx N/M$ arctan operations and a small amount of additional processing to ensure the unwrapping occurs properly. An arctan operation can be computed with polynomial approximation in approximately 5 real multiplications, a similar number of additions and 7 stored constants [14]. The polynomial curve fitting requires $\approx PN/M$ real multiplications and additions. For Step 4, there are only $O(P)$ operations. One iteration of the algorithm may be adequate to achieve good accuracy, and if not, one can use multiple iterations of the 4-step procedure. The computational

TABLE I
Computational Burden for the Filtering/Phase Unwrap Algorithm

Step	Approx. No. of Multiplies	Approx. No. of Adds
1 De-chirp	$N(4 + 16P)$	$N(2 + 16P)$
2 Filter	1	$2N$
3 Phase unwrap/regression	$N \left(\frac{5+P}{M} \right)$	$N \left(\frac{5+P}{M} \right)$
4 Adjust final estimates	P	P
5 Check if estimation successful	$N(8 + 16P) + 2$	$N(4 + 16P)$
Total for iteration, r	$DM(r) = N \left(12 + 32P + \frac{5+P}{M} \right) + P + 3$	$DA(r) = N \left(8 + 32P + \frac{5+P}{M} \right) + P$
Overall total	$\sum_r DM(r)$	$\sum_r DA(r)$

TABLE II
Computational Burden for a Newton Algorithm

	Approx. No. of Multiplications	Approx. No. of Additions
Total	$NK(4P^2 + 15P)$	$NK(4P^2 + 15P)$

burden of the overall algorithm is summarized in Table I.

By contrast, a gradient-based algorithm would typically take approximately KP^2N complex multiplications, KP^2N complex additions, and KPN exponent operations where K is a small number whose value is typically between 5 and 10 [2]. This computational burden is summarized in Table II.

IV. APPLICATION OF THE NEW ALGORITHM

A. Second-Order Polynomial Phase Examples

The first test signal considered is referred to as “signal 1.” This signal has parameters given by $P = 2$, $a_1 = \pi/2$, $a_2 = \pi/2048$, $a_0 = \pi/4$, $b_0 = 1$, and $N = 439$. Fig. 1 presents the a_2 estimate mean-square errors (MSEs) obtained for this signal with 1) a gradient-based Newton algorithm using 5 iterations (dotted line), 2) the filtering/phase unwrapping method (circles), and 3) phase unwrapping without the filtering (dashed line). The CR bounds are shown as full lines. Plots of the other MSEs show similar trends and are omitted for the sake of brevity. All methods were initiated with coarse estimates from the high-order ambiguity function (HAF) method [4] using a lag of $(N - 1)/2$. Given that the SNR threshold for the initial coarse estimation method (i.e., the second-order HAF method with $N = 439$) is known to be ≈ -3 dB, one could design the filter length, M , so that the filtering/phase unwrapping refinement method also thresholds at no greater than -3 dB. Accordingly, (10) could be used, along with the fact that $S_t \approx 9$ dB, to obtain an M value of 15. However the above approach was not actually used for the simulations. Rather, M was set initially to a small

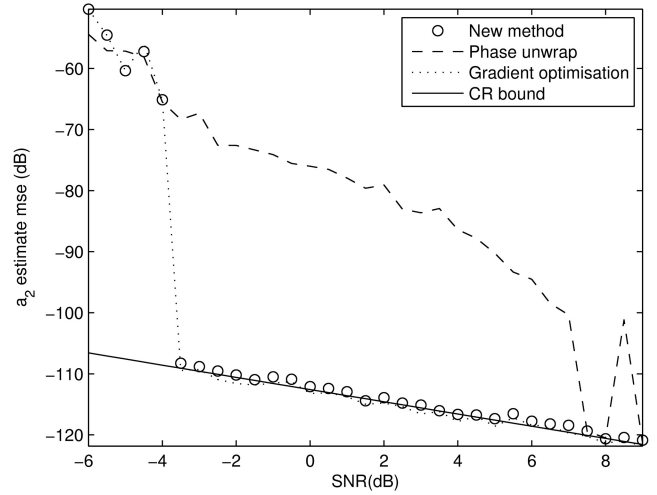


Fig. 1. a_2 estimate MSE versus SNR for signal 1.

value ($M = 5$) and iteratively increased by 4 until the estimation was gauged to be successful (as proposed in Section III). This demonstrates the effectiveness of the iterative method outlined in Section III.

There appears to be little difference between the MSEs produced by the “classical” ML-based Newton algorithm approach, the filter/phase unwrapping method, and the CR bound (as predicted in the Appendix). This observation supports the contention that the filtering/phase unwrapping method is a viable alternative to a Newton method. The phase unwrapping approach has a much higher threshold than either of the other two methods.

For the second simulation example, signal 2 was used. This signal had the same parameters as signal 1, except for the a_2 parameter, which was changed to $\pi/16$. In the simulations for this example, the initial coarse estimates were different than those used for Example 1. They were still obtained with the HAF method, but with a lag of 8 rather than $(N - 1)/2$. Using such a small lag value enables a wide range of chirp rates to be estimated [9] but causes the MSE of the (coarse initial) estimates to be relatively large. The high MSEs of the initial coarse estimates present a significant challenge for subsequent “refinement” attempts. Fig. 2 shows the MSEs obtained for 1) the

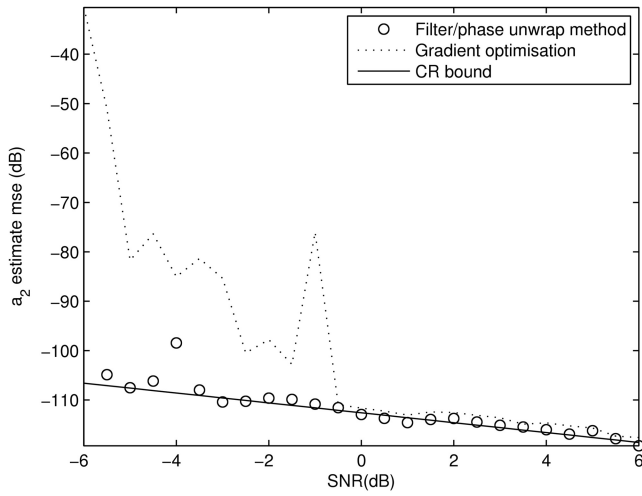


Fig. 2. a_2 estimate MSE versus SNR for signal 2.

final refined estimates obtained with the filter/phase unwrap method (circles) and 2) the final refined estimates obtained with the Newton algorithm (dotted). Again, only the a_2 estimate MSEs are shown for the sake of brevity.

The filter/phase unwrap method performs much better in refining the estimates than the Newton algorithm in this example. The filtering/phase unwrapping method has a much lower SNR threshold and produces estimates that meet the CR bound above this threshold (as predicted by the analysis in the Appendix). Note that 5 iterations were again used for the Newton algorithm.

B. Fifth-Order Polynomial Phase Example

The third signal analysed is referred to as signal 3. It is a 5th-order polynomial phase signal imbedded in white Gaussian noise. This signal has parameters given by $a_0 = 0$, $a_1 = \pi/8$, $a_2 = 1.25e-3$, $a_3 = 1.25$, $a_4 = 1e-8$, $a_5 = 1.8e-11$, $b_0 = 1$, $M = 9$, and $N = 515$. While there are many algorithms that could be used to determine the coarse initial polynomial phase parameter estimates, the one chosen is based on the “higher order phase (HP) functions” introduced in [10]. These functions yield estimates in a computationally efficient manner as outlined in [10, Section 7]. The estimation procedure is restated in the following paragraph with special attention to the case at hand.

For signal 3, the first step in determining initial polynomial phase parameter estimates is to find instantaneous frequency rate (IFR) estimates at 4 different time positions (i.e., at 4 different n values) from a 5th order “HP function” [10]. Mathematically, the IFR of a continuous time 5th-order polynomial phase signal is defined as

$$\begin{aligned} \text{IFR}(t) &= \frac{d^2 \left(\sum_{k=0}^5 a_k t^k \right)}{dt^2} \\ &= 2a_2 + 6a_3 t + 12a_4 t^2 + 20a_5 t^3. \end{aligned} \quad (12)$$

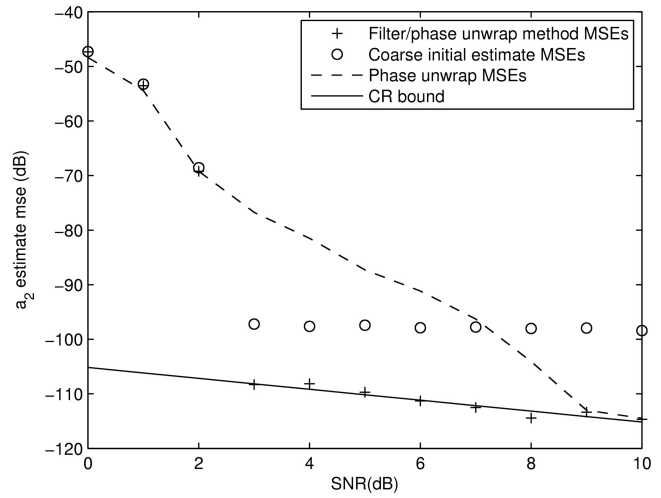


Fig. 3. a_5 estimate MSE versus SNR for signal 3.

When evaluated at integer sample points, the IFR becomes

$$\text{IFR}(n) = 2a_2 + 6a_3 n + 12a_4 n^2 + 20a_5 n^3. \quad (13)$$

The IFR estimates defined in [10] are

$$\begin{aligned} \widehat{\text{IFR}}(n) &= \arg \max_{\Omega} \sum_m z(n + 0.7629m) z(n - 0.7629m) \\ &\quad \times z(n + 0.9044m) z(n - 0.9044m) \\ &\quad \times [z(n + m) z(n - m)]^* e^{-j\Omega m^2} \end{aligned} \quad (14)$$

and $\arg \max_{\Omega} \{ \cdot \}$ indicates “the value of Ω which maximizes $\{ \cdot \}$.” The 4 time positions chosen for the IFR estimation were $n = -(N+1)/24$, $n = -(N+1)/8$, $n = (N+1)/24$ and $n = (N+1)/8$. The 4 IFR estimates found at these times were then used, along with (13), to set up four equations in four unknowns (corresponding to the parameters, a_2 , a_3 , a_4 , a_5). These four equations were then solved to produce estimates for a_2 , a_3 , a_4 , and a_5 . Then, the observation was de-chirped according to (3), leaving a near sinusoidal signal. a_1 was estimated from this signal using the standard technique of extracting the peak of the discrete-time Fourier transform.

The initial estimates obtained from the HP functions were fed into the refinement algorithm in Section II. Two iterations of the algorithm were used, as suggested in Section III, to limit amplitude distortion effects. 100 simulations were run for SNRs ranging from 0 dB to 10 dB, and the MSEs obtained for the a_5 phase parameters are shown in Fig. 3. (Only one of the parameter estimate MSEs is shown in the interests of brevity). The initial “coarse” estimate MSEs are shown as “o” symbols, the MSEs for the filter/phase unwrap method are shown as “+” symbols, while the CR bounds are shown as straight lines. The MSEs obtained with the phase unwrap method are shown as a dashed line. The MSEs for

the final parameter estimates in the new method (with $M = 9$) are quite close to the CR bounds (as predicted in the analysis in the Appendix) above a threshold of about 3 dB. The MSEs of the phase unwrapping method meet the CR bound above a threshold of about 8 dB.

The procedure for determining the CR bounds for polynomial phase signals conforming to the signal model in (1) is outlined in [17]. The formula for the CR bound for estimating the k th-order phase parameter of a P th-order polynomial phase signal conforming to the model in (1) is

$$\text{CRB}_{k,P}(N, \text{SNR}) = \frac{(\mathbf{H}_{P+1}^{-1})_{k,k}}{2\text{SNR}} \quad (15)$$

where $\mathbf{H}_{P+1}$ is a $P+1$ dimensional Hankel matrix with the element in the k th row and l th column given by $(\mathbf{H}_{P+1})_{k,l} = \sum_{n=-N/2}^{N/2} n^{k+l}$. If $N \gg 1$, then $(\mathbf{H}_{P+1})_{k,l} \approx 2(N/2)^{k+l+1}/(k+l+1)$ for $k+l$ even, and $(\mathbf{H}_{P+1})_{k,l} = 0$ for $k+l$ odd. Then, the CRB can be approximated to first order as

$$\text{CRB}_{k,P}(N, \text{SNR}) \approx \frac{(\tilde{\mathbf{H}}_{P+1}^{-1})_{k,k}}{N^{2k+1}\text{SNR}} \quad (16)$$

where $(\tilde{\mathbf{H}}_{P+1})_{k,l}$ is $1/2^{k+l}(k+l+1)$ for $k+l$ even and is 0 for $k+l$ odd.

As mentioned previously, the SNR threshold for the overall estimation procedure is limited by the threshold of the algorithm for determining the initial guesses—in this case, the HP function method. The SNR threshold is seen to be around 3 dB.

V. CONCLUSIONS

This paper has presented a statistical analysis of an alternative to a conventional Newton algorithm for refining estimates of the parameters of polynomial phase signals embedded in additive noise. This alternative is based on a de-chirping, filtering, decimation, phase unwrapping, and linear model fitting approach. A first-order statistical analysis has shown that the algorithm is asymptotically efficient, i.e., the MSEs of the parameter estimates meet the CR bound as the number of samples in the observation approaches infinity. The analysis has also shown that the “initial guesses” for parameter estimates can be significantly less accurate for the algorithm in this paper than for Newton algorithms. Simulations confirm both the results of the statistical analysis and the good performance of the algorithm.

APPENDIX. STATISTICAL ANALYSIS OF PARAMETER ESTIMATES

Consider the de-chirped signal which is created in Step 1 of the algorithm in Section II. An expression

for this signal is

$$z_d(n) = [z_s(n) + z_w(n)] \exp \left(-j \sum_{k=1}^P \hat{a}_{k_i} n^k \right) \quad (17)$$

$$= z_{ds}(n) + z_{dw}(n) \quad (18)$$

where

$$z_{dw}(n) = z_w(n) \exp \left(-j \sum_{k=1}^P \hat{a}_{k_i} n^k \right) \quad (19)$$

$$z_{ds}(n) = b_0 e^{ja_0} \exp \left(j \sum_{k=1}^P \delta a_k n^k \right) \quad (20)$$

and $\delta a_k = a_k - \hat{a}_{k_i}$ for $k = 1, \dots, P$. The de-chirped signal, $z_d(n)$, as seen in (18), consists of two components. The first, $z_{ds}(n)$, is a polynomial phase signal, with polynomial phase parameters, $\{a_0, \delta a_1, \delta a_2, \dots\}$. If the initial estimates are reasonably accurate, the spectral content of this component would be localized in the baseband region, i.e., around 0 Hz. (More will be said on the required accuracy of the initial estimates later in the Appendix). The second component $z_{dw}(n)$ is a random noise term, whose covariance is given by

$$C_{z_{dw}}(n_1, n_2) = E\{z_{dw}(n_1)z_{dw}^*(n_2)\} \quad (21)$$

$$= E \left\{ \exp \left(-j \sum_{k=1}^P \hat{a}_{k_i} n_1^k \right) z_w(n_1) \times \exp \left(j \sum_{k=1}^P \hat{a}_{k_i} n_2^k \right) z_w^*(n_2) \right\} \quad (22)$$

$$= E \left\{ \exp \left(j \sum_{k=1}^P \hat{a}_{k_i} (n_2^k - n_1^k) \right) z_w(n_1) z_w^*(n_2) \right\}. \quad (23)$$

Now if it is assumed that

$$E \left\{ \exp \left(j \sum_{k=1}^P \hat{a}_{k_i} (n_2^k - n_1^k) \right) z_w(n_1) z_w^*(n_2) \right\} \approx E\{z_w(n_1)z_w^*(n_2)\} = \sigma^2 \delta(n_1 - n_2) \quad (24)$$

then

$$C_{z_{dw}}(n_1, n_2) \approx \sigma^2 \delta(n_1 - n_2) \quad (25)$$

where $\delta(n)$ denotes the Kronecker delta function. If (24) is a reasonable approximation, then $C_{z_{dw}}$ has an approximately diagonal structure, with each of the elements on the diagonal being identical. The $z_{dw}(n)$ noise term is therefore white noise with variance equal to σ^2 . It is important to comment on the “reasonableness” of the assumption specified in (24). This assumption is quite viable for a variety of practical scenarios, as explained below.

Recall that $\hat{a}_{k_i} = a_k - \delta a_k$. Then, the left-hand side of (24) may be rewritten as

$$\begin{aligned} & E \left\{ \exp \left(j \sum_{k=1}^P \hat{a}_{k_i} (n_2^k - n_1^k) \right) z_w(n_1) z_w^*(n_2) \right\} \\ &= \exp \left(j \sum_{k=1}^P a_k (n_2^k - n_1^k) \right) \\ & \times E \left\{ \exp \left(-j \sum_{k=1}^P \delta a_k (n_2^k - n_1^k) \right) z_w(n_1) z_w^*(n_2) \right\}. \end{aligned} \quad (26)$$

In many polynomial phase estimation algorithms, δa_k is $O(N^{-k-1/2})$ [4, 10] so $\sum_{k=1}^P \delta a_k (n_2^k - n_1^k)$ is $O(N^{-1/2})$ or less. Then $\exp(-j \sum_{k=1}^P \delta a_k (n_2^k - n_1^k)) \approx 1$ and (26) becomes

$$\begin{aligned} & E \left\{ \exp \left(j \sum_{k=1}^P \hat{a}_{k_i} (n_2^k - n_1^k) \right) z_w(n_1) z_w^*(n_2) \right\} \\ & \approx \exp \left(j \sum_{k=1}^P a_k (n_2^k - n_1^k) \right) E \{ \exp(1 \cdot z_w(n_1) z_w^*(n_2)) \} \end{aligned} \quad (27)$$

$$\approx \exp \left(j \sum_{k=1}^P a_k (n_2^k - n_1^k) \right) E \{ \exp(z_w(n_1) z_w^*(n_2)) \} \quad (28)$$

$$\approx 0 \quad \text{if } n_1 \neq n_2 \quad \text{and} \quad \sigma^2 \quad \text{if } n_1 = n_2. \quad (29)$$

The above result shows that (24) is approximately true if δa_k is $O(N^{-k-1/2})$. Even where δa_k is not $O(N^{-k-1/2})$, the assumption in (24) can often be quite reasonable. To appreciate this, consider the following. Suppose an alternative coarse initial estimate $\hat{a}_{k_1}$ is formed in the same way that $\hat{a}_{k_i}$ is formed, except that the noise on the two samples at n_1 and n_2 (only) are set to 0, i.e., the noise on the samples at n_1 and n_2 do not influence the estimate, $\hat{a}_{k_1}$. Also let the error in the $\hat{a}_{k_1}$ estimate be denoted by $\delta a_{k_1} = a_{k_1} - \hat{a}_{k_1}$. Now using the fact that $\delta a_k = \delta a_{k_1} + (\delta a_k - \delta a_{k_1})$ and $\hat{a}_{k_i} = a_k - \delta a_k$, (23) can be rewritten as

$$\begin{aligned} & C_{z_{dw}}(n_1, n_2) \\ &= E \left\{ \exp \left(j \sum_{k=1}^P (a_k - \delta a_{k_1} - (\delta a_k - \delta a_{k_1})) (n_2^k - n_1^k) \right) \right. \\ & \quad \left. \times z_w(n_1) z_w^*(n_2) \right\} \\ &= \exp \left(j \sum_{k=1}^P (a_k - \delta a_{k_1}) (n_2^k - n_1^k) \right) \\ & \times E \left\{ \exp \left(-j \sum_{k=1}^P (\delta a_k - \delta a_{k_1}) (n_2^k - n_1^k) \right) z_w(n_1) z_w^*(n_2) \right\} \end{aligned} \quad (30)$$

since δa_{k_1} is independent of $z_w(n_1)$ and $z_w(n_2)$. One would typically expect the zeroing of the noise in two samples from an asymptotically long observation record to make an extremely small difference in the estimate obtained, i.e., one would typically expect that $\delta a_{k_1} - \delta a_k \approx 0$. For this reason, $\exp(-j \sum_{k=1}^P (\delta a_k - \delta a_{k_1}) (n_2^k - n_1^k))$ will often be ≈ 1 . Then $C_{z_{dw}}(n_1, n_2) \approx 0$ if $n_1 \neq n_2$ and $C_{z_{dw}}(n_1, n_2) \approx \sigma^2$ if $n_1 = n_2$, i.e., as long as it can be assumed that $(\delta a_k - \delta a_{k_1}) (n_2^k - n_1^k) \approx 0$, then $C_{z_{dw}}(n_1, n_2) \approx \sigma^2 \delta(n_1 - n_2)$.

In Step 2 of the algorithm, $z_d(n)$ is subjected to the filtering and decimation procedure defined in (4). The resulting output is

$$z_o(m) = \frac{1}{M} \sum_{n=mM-(M-1)/2}^{mM+(M-1)/2} [z_{ds}(n) + z_{dw}(n)] \quad (31)$$

$$= z_{os}(m) + z_{ow}(m) \quad (32)$$

where

$$z_{os}(m) = \frac{1}{M} \sum_{n=mM-(M-1)/2}^{mM+(M-1)/2} z_{ds}(n) \quad (33)$$

and

$$z_{ow}(m) = \frac{1}{M} \sum_{n=mM-(M-1)/2}^{mM+(M-1)/2} z_{dw}(n). \quad (34)$$

Because $z_{ow}(m)$ is the summation of M samples of white Gaussian noise with variance σ^2 , scaled by $1/M$, it will be white Gaussian noise with variance σ^2/M . The “signal component” $z_{os}(m)$ can be expressed as

$$z_{os}(m) = \frac{1}{M} \sum_{n=mM-(M-1)/2}^{mM+(M-1)/2} b_0 e^{ja_0} \exp \left(j \sum_{k=1}^P \delta a_k n^k \right) \quad (35)$$

$$\begin{aligned} &= \frac{b_0 e^{ja_0}}{M} \sum_{v=-(M-1)/2}^{(M-1)/2} \exp \left(j \sum_{k=1}^P \delta a_k (mM + v)^k \right) \\ &= \frac{b_0 e^{ja_0}}{M} \sum_{v=-(M-1)/2}^{(M-1)/2} \exp(j\psi(m)) \end{aligned} \quad (36)$$

where

$$\psi(m) = \sum_{k=1}^P [\delta a_k (mM)^k + k \delta a_k (mM)^{k-1} v + \dots + \delta a_k v^k]. \quad (37)$$

Before proceeding any further, it is necessary to consider the required accuracy of the initial estimates $\hat{a}_{k_i}$ which are fed to the refinement algorithm in Section II. If the initial coarse estimates were all perfectly accurate, the change in phase

between successive samples in the de-chirped signal would be 0. In practice, however, there will be inaccuracies in the initial estimates that will give rise to non-zero phase changes between samples in the de-chirped signal. It is important to constrain the extent of these changes, such that after de-chirping/filtering/decimation, phase unwrapping can still be easily achieved. Accordingly, it will be required that the maximum consecutive change in phase in the final decimated signal be less than $\pi/4$ in magnitude. The following shows that this can be achieved, provided that the following constraints are applied to the initial coarse estimates:

$$|\delta a_k| < \frac{\pi}{4kPM(N/2)^{k-1}}, \quad k = 1, \dots, P. \quad (38)$$

If the above constraints are imposed, then the maximum consecutive change in phase in the de-chirped signal due to the error in $\hat{a}_{k_i}$ is given by

$$|\text{Phase_change_max_de-chirped}_k| = \delta a_k (n^k - n^{k-1})|_{n=(N-1)/2} \quad (39)$$

$$\approx \delta a_k k(N/2)^{k-1}. \quad (40)$$

Substituting (38) into the above leads to

$$|\text{Phase_change_max_de-chirped}_k| \approx \frac{\pi k(N/2)^{k-1}}{4kPM(N/2)^{k-1}} \quad (41)$$

$$= \frac{\pi}{4PM}. \quad (42)$$

Then, the maximum consecutive change in phase in the de-chirped signal due to the errors in all P of the $\hat{a}_{k_i}$ is given by

$$|\text{Phase_change_max_de-chirped}_{\text{total}}| \approx P \frac{\pi}{4PM} \quad (43)$$

$$= \frac{\pi}{4M}. \quad (44)$$

Since the de-chirped signal is subsequently filtered and decimated by M , the maximum consecutive change in phase in the decimated signal (assuming minimal distortion of the signal by the filtering) is given by

$$|\text{Phase_change_max_decimated}_{\text{total}}| = M \frac{\pi}{4M} = \frac{\pi}{4}. \quad (45)$$

The following shows that the filtering introduces minimal distortion to the signal, provided that the constraints in (38) are maintained. This paper assumes that

$$M \ll N \quad \text{and} \quad P \ll N. \quad (46)$$

Also, since $|v|$ is always less than M ,

$$|v| \ll N. \quad (47)$$

It is now pertinent to find an approximate expression for $\psi(m)$ in (37). From (38), it is clear that $|\delta a_k|$ is $O(N^{-(k-1)})$ or less. Using (46) and (47) and neglecting all but first-order terms, the expression for $\psi(m)$ in (37) can be approximated as

$$\psi(m) = \sum_{k=1}^P [\delta a_k (mM)^k + k \delta a_k (mM)^{k-1} v]. \quad (48)$$

Substituting this result into (36) gives

$$\begin{aligned} z_{os}(m) &\approx \frac{b_0 e^{ja_0}}{M} \sum_{v=-(M-1)/2}^{(M-1)/2} \exp \left(\sum_{k=1}^P j \delta a_k (mM)^k \right) \\ &\times \exp \left(\sum_{k=1}^P j k \delta a_k (mM)^{k-1} v \right) \\ &= \frac{b_0 e^{ja_0}}{M} \exp \left(\sum_{k=1}^P j \delta a_k (mM)^k \right) \\ &\times \sum_{v=-(M-1)/2}^{(M-1)/2} \exp \left(\sum_{k=1}^P j k \delta a_k (mM)^{k-1} v \right). \end{aligned} \quad (49)$$

Now to simplify notation, the following assignment is made:

$$\Theta = \sum_{k=1}^P k \delta a_k (mM)^{k-1}. \quad (51)$$

Then the above equation can be reexpressed as

$$\begin{aligned} z_{os}(m) &\approx \frac{b_0 e^{ja_0}}{M} \exp \left(\sum_{k=1}^P j \delta a_k (mM)^k \right) \\ &\times \sum_{v=-(M-1)/2}^{(M-1)/2} \exp(j\Theta v) \end{aligned} \quad (52)$$

$$\begin{aligned} &= \frac{b_0 e^{ja_0}}{M} \exp \left(\sum_{k=1}^P j \delta a_k (mM)^k \right) \\ &\times \left(\frac{\exp(j\Theta)^{-(M-1)/2} - \exp(j\Theta)^{(M+1)/2}}{1 - \exp(j\Theta)} \right) \end{aligned} \quad (53)$$

$$\begin{aligned} &= \frac{b_0 e^{ja_0}}{M} \exp \left(\sum_{k=1}^P j \delta a_k (mM)^k \right) \\ &\times \left(\frac{[\exp(j\Theta/2)][\exp(-jM\Theta/2) - \exp(jM\Theta/2)]}{\exp(-j(\pi/2 - \Theta/2)) \sqrt{(1 - \cos(\Theta))^2 + (\sin(\Theta))^2}} \right). \end{aligned} \quad (54)$$

Now using (38) and the fact that $mM < N/2$, it follows that

$$\Theta < \frac{\pi}{4M}. \quad (55)$$

The smallest nontrivial value for M is 3 (since M must be odd) and so Θ is at most $\pi/12$. Θ is therefore

a small angle. Equation (54) can therefore be rewritten as

$$z_{os}(m) \approx \frac{b_0 e^{ja_0}}{M} \exp \left(\sum_{k=1}^P j \delta a_k (mM)^k \right) \times \left(\frac{\exp(-jM\Theta/2) - \exp(jM\Theta/2)}{\exp(-j\pi/2) \sin(\Theta)} \right) \quad (\text{since } \Theta \text{ is small}) \quad (56)$$

$$= \frac{b_0 e^{ja_0}}{M} \exp \left(\sum_{k=1}^P j \delta a_k (mM)^k \right) \times \left(\frac{\exp(jM\Theta/2) - \exp(-jM\Theta/2)}{j \sin(\Theta)} \right) \quad (57)$$

$$\approx \frac{b_0 e^{ja_0}}{M} \exp \left(\sum_{k=1}^P j \delta a_k (mM)^k \right) \times \left(\frac{\exp(jM\Theta/2) - \exp(-jM\Theta/2)}{j(\Theta)} \right) \quad (58)$$

$$= \frac{b_0 e^{ja_0}}{M} \exp \left(\sum_{k=1}^P j \delta a_k (mM)^k \right) M \text{sinc}(M\Theta/2) \quad (59)$$

$$= b_0 e^{ja_0} \exp \left(\sum_{k=1}^P j \delta a_k (mM)^k \right) \text{sinc}(M\Theta/2) \quad (60)$$

$$\approx b_0 e^{ja_0} \exp \left(\sum_{k=1}^P j \delta a_k (mM)^k \right) \quad (61)$$

where $\text{sinc}(x) = \sin(x)/x$. Note that the approximation in (61) is possible because, from (55), $M\Theta/2 < \pi/8$, and so $\text{sinc}(M\Theta/2) \approx 1$; see (55).

Comparison of (61) and (20) indicates that $z_{os}(m)$ can be approximated as a polynomial phase signal that is a time-scaled version of $z_{ds}(n)$. That is, the filtering introduces minimal distortion to the signal content. $z_{os}(m)$ has an amplitude of b_0 and has $\approx N/M$ samples. $z_{ow}(n)$ is white Gaussian noise with variance σ^2/M . Therefore, $z_o(m)$ (the sum of $z_{os}(m)$ and $z_{ow}(m)$) is, to first-order approximation, an N/M sample polynomial phase signal in white Gaussian noise with an SNR of Mb_0^2/σ^2 . At high SNR, phase unwrapping and linear least-squares estimation produce statistically efficient estimates for polynomial phase signals [16, 6]. Thus, the phase parameters δa_k estimated in Step 3 of the algorithm will have variances that meet the CR bound for asymptotically long signals, i.e., the k th-order phase parameter will have an asymptotic variance of

$$\text{var}(\hat{\delta a}_k) \approx \text{CRB}_{k,P}(N/M, Mb_0^2/\sigma^2). \quad (62)$$

In the above, $\text{CRB}_{k,P}(\text{NUM}, \text{SNR})$ denotes the CR lower variance bound for a k th-order polynomial

phase parameter, assuming the observation conforms to the following signal model:

$$z_o(m) = \text{AMP} \cdot \exp \left\{ j \sum_{k=0}^P \delta a_k m^k \right\} + z_{ow}(m), \quad |m| \leq \frac{\text{NUM} - 1}{2} \quad (63)$$

where AMP is the signal amplitude, P is the polynomial phase order, the $\{\delta a_k, k = 0, \dots, P\}$ are the phase parameters to be estimated, the sampling rate is unity, and $z_{ow}(n)$ is additive complex white Gaussian noise of power AMP^2/SNR .

The variance of the final a_k estimate can be obtained from the the variance of the δa_k estimate in (62) by using the relation in (7). The variance of the final a_k estimate is

$$\text{var}(\hat{a}_{k_f}) \approx M^{-(2k)} \cdot \text{CRB}_{k,P}(N/M, Mb_0^2/\sigma^2) \quad (64)$$

$$= M^{-(2k+1)} \cdot \text{CRB}_{k,P}(N/M, b_0^2/\sigma^2) \quad (65)$$

$$= \text{CRB}_{k,P}(N, b_0^2/\sigma^2). \quad (66)$$

Equation (66) follows from (65) because the CR bound of the k th-order polynomial phase parameter is proportional to the number of samples raised to the $-(2k+1)$ th power [17]. (66) indicates that the variance of the final estimate of the k th polynomial phase parameter is approximately equal to the CR lower variance bound, assuming N samples in the observation and an SNR of b_0^2/σ^2 , i.e., assuming the original signal model. The approximation approaches exactness in the limit as $N \rightarrow \infty$.

A. The Required Accuracy of the Initial Estimates

Note that (38) is effectively a statement of the required accuracy for the initial estimates for the new algorithm. In general, the required accuracy is substantially lower than for an equivalent Newton algorithm. For example, if the observation is a sinusoid, (38) indicates that the initial estimate for the frequency needs to be accurate enough to guarantee that

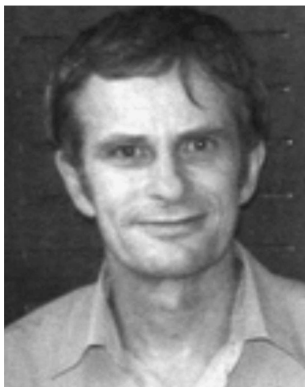
$$|\delta a_1| < \frac{\pi}{4M}. \quad (67)$$

The requirement of the initial estimate error is therefore only an $O(1)$ requirement. By contrast, a Newton algorithm would require the initial estimate error to be $O(N^{-3/2})$ [1, 2].

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